

TA 8.7 Damping-Factor-Control Frequency Compensation Technique for Low-Voltage Low-Power Large Capacitive Load Applications

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Frequency compensation techniques for multiple-stage amplifiers are becoming increasingly important as cascode configuration is not applicable to low-voltage design. Nested Miller compensation (NMC) is commonly used to stabilize multiple-stage amplifiers. However, the bandwidth of an NMC amplifier is poor. Several topologies such as multi-path nested Miller compensation (MNM) and nested Gm-C compensation (NGCC) are proposed to enhance the bandwidth [1, 2, 3]. When compared to an NMC amplifier, MNM can increase the bandwidth by approximately a factor of two while NGCC can further improve the stability of the amplifier. Nevertheless, the bandwidth enhancement by the two topologies is not sufficient for high-speed applications especially to drive large capacitive loads, such as the error amplifier in a low-voltage low-dropout regulator in portable electronic devices. A topology called damping-factor-control frequency compensation (DFCFC) for three-stage amplifiers significantly increases the bandwidth and improves the transient response of the amplifiers [4].

In a three-stage NMC amplifier, the gain-bandwidth product (GBW) is reduced by the inner compensation capacitor [2]. If only the outer compensation capacitor is used, the bandwidth can be maximized. However, the amplifier is unstable because the damping factor of the two high frequency complex poles is too small, and a frequency 'peak', which causes instability, is generated. Therefore, additional circuitry is needed to stabilize the amplifier by controlling the damping factor of the second and third pole. The structure to implement the above compensation technique is shown in Figure 8.7.1. The gain stage $-A_{v1}$ is the extra circuitry which is used to control the damping factor of the second and third pole. In addition, a feed-forward transconductance stage in this topology forms a push-pull output stage which reduces distortion of the output signal caused by Class-A output stage. With $C_{m1}=C_{m2}$, $A_{v1} \gg 1$ and negligible C_{p1} and C_{p2} , the corresponding small-signal voltage transfer function is given by:

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{A_{DC} \left(1 + s \frac{C_{p2}g_{m2} - C_{m1}g_{m4}}{g_{m2}g_{m3} + g_{mf2}g_{m4}} - s^2 \frac{C_{p2}C_{m1}}{g_{m2}g_{m3} + g_{mf2}g_{m4}} \right)}{\left(1 + \frac{s}{p_{-3dB}} \right) \left(1 + s \frac{C_Lg_{m4}}{g_{m2}g_{m3} + g_{mf2}g_{m4}} + s^2 \frac{C_{p2}C_L}{g_{m2}g_{m3} + g_{mf2}g_{m4}} \right)}$$

$$\text{where } A_{DC} = g_{m1}g_{m2}g_{m3}R_1R_2R_3 \text{ and } p_{-3dB} = (C_{m1}g_{m2}g_{m3}R_1R_2R_3)^{-1}. \quad (1)$$

From (1), it can be seen that the second-order function in the denominator is controlled by g_{m4} . Thus, stability of the amplifier can be controlled with an optimum value of g_{m4} . Moreover, as the second and third pole of the system are dependent of C_{p2} , the poles are located at higher frequencies, and a larger GBW is obtained.

The stability of this topology is achieved by considering that the poles of the amplifier in unity-feedback configuration have third-order Butterworth response. Setting g_{m2} equal to g_{m3} , the stability criteria are given by:

$$C_{m1} = C_{m2} = \frac{4}{K} \cdot \left(\frac{g_{m1}}{g_{m3}} \right) \cdot C_L \text{ and } g_{m4} = K \cdot \left(\frac{C_{p2}}{C_L} \right) \cdot g_{m3} \quad (2)$$

where $K = \sqrt{1 + 2(C_L / C_{p2})(g_{m2} / g_{m3})}$. From (2), the values of C_{m1} , C_{m2} and g_{m4} will be smaller for a smaller C_{p2} to C_L ratio. The area of the layout occupied by the compensation capacitors is therefore reduced when compared to the NMC amplifier, and there is only a small increase in power consumption for small g_{m4} . Moreover, the zeros of the system are located at high frequencies since C_{m1} , C_{p2} and g_{m4} are small.

Applying the stability conditions in (2) to (1), the GBW of the DFCFC amplifier is expressed by:

$$GBW_{(DFCFC)} = K \cdot \left(\frac{g_{m3}}{4C_L} \right) = K \cdot GBW_{(NMC)} \quad (3)$$

The GBW is enhanced by a factor of K compared to that of the NMC amplifier [1, 2]. If C_{p2} is much smaller than C_L , the increase in the GBW is large. Therefore, this topology is suitable for amplifiers driving large capacitive loads. As the dimension conditions in (2) are dependent on the ratios of the capacitances and transconductances, the stability of the DFCFC amplifier is less sensitive to the global variations of circuit parameters.

The slew rate of a Miller-compensated amplifier is I_{input}/C_{m1} where I_{input} is the biasing current of the differential input stage. As C_{m1} is reduced by a factor of K, the slew rate of the DFCFC amplifier is increased by K times, and a faster transient response is obtained.

The transistor level realization and the micrograph of the DFCFC amplifier are shown in Figure 8.7.2 and Figure 8.7.3, respectively. Amplifiers using DFCFC and the conventional NMC are fabricated using the AMS (Austria Mikro Systeme Group, Austria) 0.8μm CMOS process. A summary of the performance of both amplifiers is listed in Table 8.7.1. For the DFCFC amplifier driving a 1nF load, the AC and transient responses of the DFCFC amplifier are shown in Figure 8.7.4 and Figure 8.7.5, respectively. With only 6.5% increase in power consumption, the GBW is increased by 18 times, the slew rate is enhanced by more than 14 times, and the settling time is reduced by approximately 9 times. The compensation capacitors of the DFCFC amplifier can be integrated, but those used in the NMC amplifier cannot. Moreover, experimental results show that the power supply rejection ratio is improved by more than 55dB at 10kHz using DFCFC.

Both theoretical analysis and experimental results show that DFCFC significantly improves bandwidth and transient performance of a three-stage amplifier with no significant increase in power consumption. This compensation technique is useful for amplifiers with large capacitive load for low-voltage high-current applications in portable electronic equipment such as the error amplifier in a low-dropout regulator.

Acknowledgment:

This work was supported by the RGC Competitive Earmarked Research Grant, Hong Kong SAR Government, HKUST6007/97E.

References:

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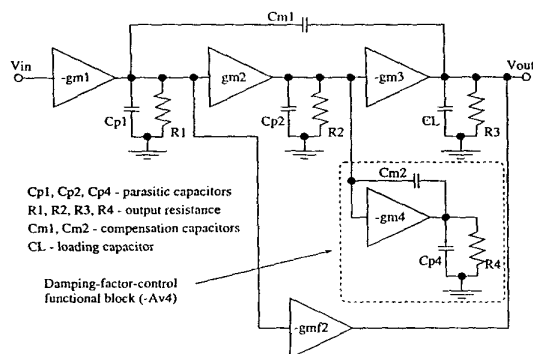


Figure 8.7.1: DFCFC topology.

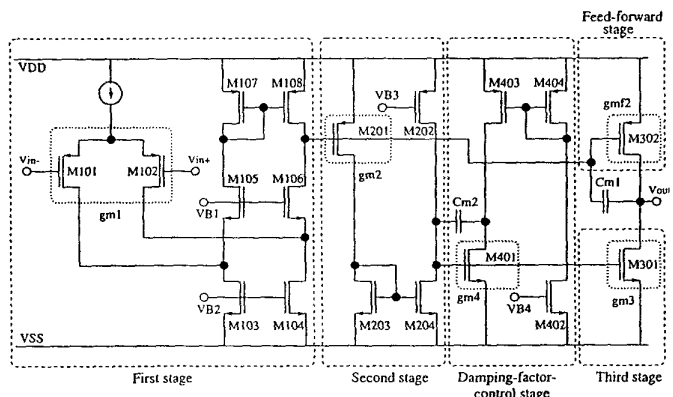


Figure 8.7.2: DFCFC amplifier circuit diagram.

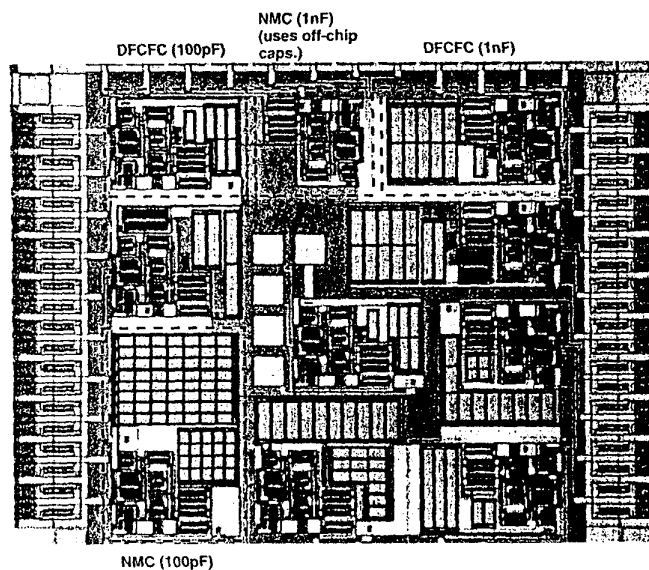


Figure 8.7.3: Micrograph of 3-stage amplifiers.

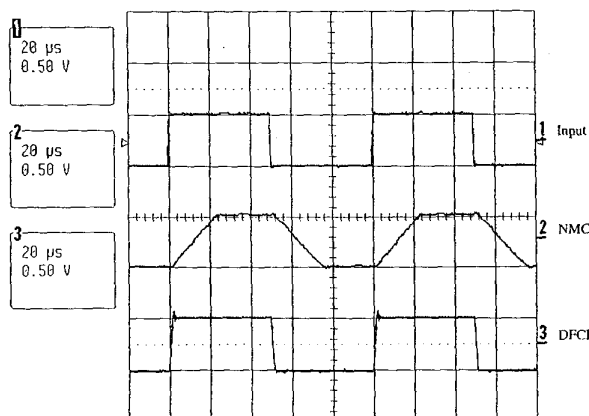


Figure 8.7.5: Amplifier transient response.

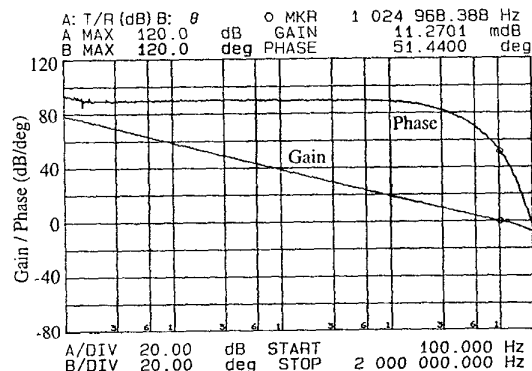


Figure 8.7.4: DFCFC amplifier AC response.

	NMC	DFCFC	NMC	DFCFC
Loading Condition	100pF/25k Ω		1nF/25k Ω	
DC Gain	>100dB	>100dB	>100dB	>100dB
GBW	590kHz	2.6MHz	55kHz	1MHz
PM	43°	43°	56°	51°
Slew Rate SR+/SR- (V/ μ s)	0.23/0.23	1.36/1.27	0.023/0.022	0.33/0.39
Settling Time Ts+/Ts- (μ s)	4.25/4.36	0.96/1.37	32.02/32.92	3.54/3.55
PSSR+ @10kHz	64.10dB	91.93dB	24.49dB	80.63dB
PSSR- @10kHz	35.61dB	82.55dB	1.31dB	65.39dB
Power Consumption	400 μ W	420 μ W	400 μ W	426 μ W
Power Supply	+/- 1V		+/- 1V	
Area	0.23mm ²	0.11mm ²	0.08mm ² (Uses off-chip caps.)	0.14mm ²

Table 8.7.1: Measured amplifier performance.